FIELD KNOWLEDGE

LLM Infrastructure Primer

The industry concepts every Mubit engineer, solutions engineer and product person needs, from APIs to adapters to MCP.

Version	Owner	Effective from	Next review	Applies to
1.2	Models Team	1 Oct 2026	1 Jan 2027	All technical and customer-facing roles

1. OpenAI-compatible APIs and base URLs

Most LLM providers accept the same request shape as OpenAI's Chat Completions API. An application can switch provider, or route through a gateway such as Mubit, by changing the base URL and API key in its SDK client. This is why connecting to Mubit can be a one-line change.

2. Tokens, context windows and cost

- Models read and write text as tokens, roughly three-quarters of an English word each.
- The context window is the maximum number of tokens in one request, including the prompt and the answer.
- Providers usually charge separately for input and output tokens, with output tokens costing more.

3. Latency

- Time to first token (TTFT) is how long before the first token streams back. It decides how fast a product feels.
- p50 is the median request; p95 is the slowest 5%. Customers notice p95.

4. Fine-tuning and adapters (LoRA)

Low-Rank Adaptation (LoRA) freezes a base model's weights and trains small low-rank matrices added to some layers. The result, an adapter, is typically megabytes rather than gigabytes. Many adapters can be served on one base model and swapped per request, which is what makes a dedicated model per task affordable.

5. Evaluation

- **Held-out set:** examples kept out of training, used only to measure. Evaluating on training data overstates quality.
- **Paired comparison:** run the current and candidate versions on the same tasks and compare.
- **Verified outcomes:** prefer tests, application checks and human review over a model grading its own work.
- **Cost per successful task:** total cost divided by tasks that actually succeeded, not by all requests.

6. Structured outputs

Many APIs can constrain a model to return JSON matching a schema. Always validate the result in code, for example with Zod or Pydantic, because a valid shape does not mean correct content.

7. Retrieval-augmented generation (RAG)

RAG retrieves relevant chunks from a document store and puts them in the prompt, so the model answers from your data and can cite it. Most answer errors in RAG systems come from retrieving the wrong chunks, not from the model.

8. Model Context Protocol (MCP)

MCP is an open protocol, introduced by Anthropic in 2024, for connecting AI applications to tools and data in a standard way.

Part	What it is	Example
Host	The AI application the user works in	A chat app or an IDE agent
Client	The connector inside the host, one per server	The host's link to a Jira server
Server	A program that exposes capabilities to hosts	A Jira or Postgres MCP server

Servers offer three kinds of primitive:

- **Tools:** actions the model can call, such as create_ticket.
- **Resources:** data the application can read into context, such as a file or a ticket.
- **Prompts:** reusable templates the user can invoke.

Local servers usually talk over stdio; remote servers use HTTP. MCP Apps, an official extension, lets a tool return an interactive interface that the host renders in a sandboxed frame.

9. Claim verification

A verification model checks a claim against a policy and the available evidence. A useful verifier says when the evidence is insufficient instead of guessing. Banyan Sapling returns supported, violation, missing information or inconsistent context.